The Journal of Open Source Software

VHF Morse Code Transmission via Display Pixel-Clock Harmonics and EMI Leakage

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 24 October 2026
Published: unpublished

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Summary

Modern video pipelines are high-speed digital systems that generate unintentional electromagnetic radiation (EMI) during normal operation[cite: 1]. The **VHF Monitor RF Transmitter** is a browser-based experimental and educational tool that converts a standard computer display into a short-range Continuous Wave (CW) and Amplitude Modulated (AM) Morse code transmitter in the VHF spectrum without requiring external radio hardware, SDRs, or modems[cite: 1]. By controlling pixel output via an HTML5 Canvas rendering engine, the software manipulates the display's video processing pipeline and transmission interfaces (HDMI/DisplayPort) to radiate modulated electromagnetic signals through physical side-channel emissions[cite: 1].

Statement of Need

Electromagnetic side-channel research (commonly known as TEMPEST) demonstrates how unintentional radio-frequency emanations from computing equipment can be exploited for data exfiltration or covert communication (Guri et al., 2014; Kuhn & Anderson, 1998). However, reproducing these phenomena in educational or laboratory settings often requires specialized hardware, custom FPGA designs, or platform-specific kernel modifications.

The **VHF Monitor RF Transmitter** addresses this limitation by providing a zero-installation, cross-platform web application capable of generating reproducible RF harmonics directly from a standard web browser[cite: 1]. Highlighted by the UK National Cyber Security Centre (NCSC), Hackaday, and RTL-SDR.com, the tool serves security researchers, educators, and amateur radio operators investigating physical-layer security and video EMI leakage. It also includes automatic localization across six languages (English, Turkish, French, German, Spanish, and Chinese), dynamic refresh-rate probing via high-resolution timers, and built-in photosensitive epilepsy safety warnings[cite: 1].

Physics and Engineering of Display EMI

To display a stable image, a graphics card pushes video data frame-by-frame governed by the **Pixel Clock Frequency** (f_{pixel}), calculated from total frame dimensions (active pixels plus horizontal and vertical blanking intervals) and the refresh rate (f_{refresh}):

$$f_{\text{pixel}} = H_{\text{total}} \times V_{\text{total}} \times f_{\text{refresh}}$$

For a standard 1080p display at 60 Hz following the CEA-861 timing standard ($H_{\text{total}} = 2200$, $V_{\text{total}} = 1125$):

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$$f_{\text{pixel}} = 2200 \times 1125 \times 60 = 148.500 \text{ MHz}$$

When the software renders an alternating 1×1 black (`#000000`) and white (`#FFFFFF`) pixel pattern, every adjacent clock cycle forces the display interface serializer (TMDS/FRL in HDMI or Main Link Lanes in DisplayPort) to toggle its voltage state rapidly[cite: 1]. This induces a carrier wave at f_{pixel} (and its higher harmonics $2f_{\text{pixel}}$, $3f_{\text{pixel}}$). By switching between the high-contrast alternating pattern (Carrier ON) and a solid black canvas (Carrier OFF) according to standardized Morse code timings, nearby VHF receivers or SDRs can demodulate audible CW/AM tones[cite: 1].

Resolution	Refresh Rate (Hz)	Total Timings ($H_{\text{total}} \times V_{\text{total}}$)	Fundamental Target Frequency
1920 x 1080	60 Hz	2200×1125	**148.500 MHz** (VHF Amateur 2m Band)
1920 x 1080	144 Hz	2200×1125	**356.400 MHz** (UHF Band)
2560 x 1440	60 Hz	2720×1481	**241.500 MHz** (VHF High)
1280 x 720	60 Hz	1650×750	**74.250 MHz** (VHF Low / 4m Band)

References

Guri, M., Kedma, G., Kachlon, A., & Elovici, Y. (2014). AirHopper: Bridging the air-gap between isolated networks and mobile phones using radio frequencies. *2014 9th IEEE International Conference on Malicious and Unwanted Software: The Americas (MALWARE)*, 58–67. https://doi.org/10.1109/MALWARE.2014.6999418

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